

Augmenting Intra-Class Difference Limb Diversity for Variety 3D Human Motions Generation

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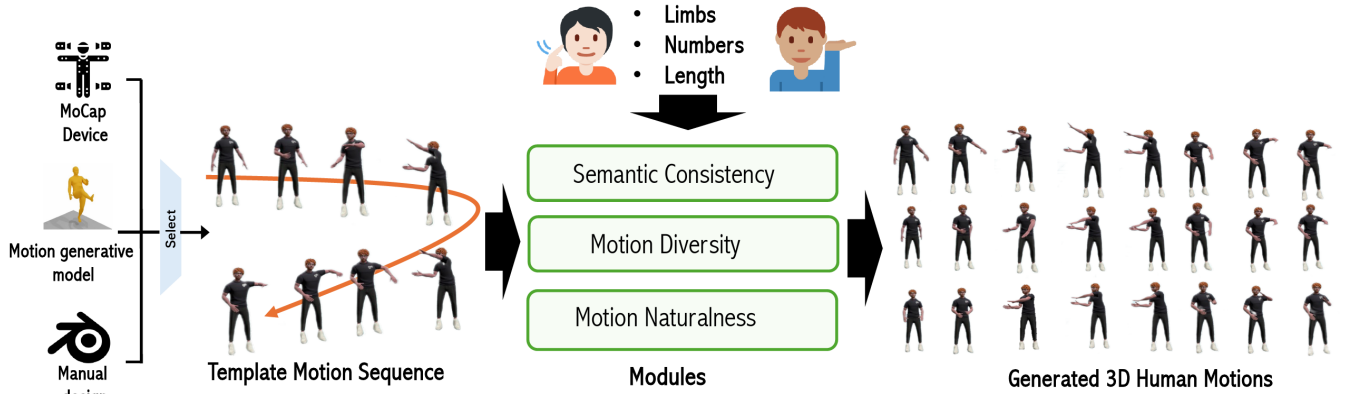


Figure 1: We present a 3D human motion generation pipeline based on motion template sequence input as a new tool. It is able to accept multi-source motion sequences, such as text-driven motion generation models and motion capture devices. It focuses on motion's intra-class difference augmentation and generates new 3D human motions with great diversity and naturalness. The tool has significant advantages in forming a lightweight and customized 3D human motion generation with user indication.

Abstract

3D human motion generation holds significant value in fields such as game development and content creation. However, existing 3D human motion generation methods often overlook intra-class difference limb movement variations during generation, resulting in insufficient limb diversity in generated motions. We show a novel pipeline, a limb intra-class difference augmentation 3D human motion generation that preserves motion semantics while amplifying the body limb diversity. This work can augment the 3D motion sequences input by the user, generating diverse new 3D motion sequences, thereby providing more possibilities for the user's creation.

CCS Concepts

• Human-centered computing → User interface toolkits.

Keywords

Motion Generation, 3D human, Human activity recognition, Intra-class difference

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1 Introduction

Creating diverse and natural 3D character animations is crucial for applications like interactive media and virtual reality [10, 14]. Current approaches generally fall into two categories. While generative models can automatically synthesize motions from various inputs, such as text descriptions or motion labels [8, 11], they often produce generic results that lack personalization and fine-grained control. On the other hand, interactive editing tools [5] provide animators with direct control, but creating rich variations can be a labor-intensive manual process.

A key limitation shared across these methods is their tendency to overlook intra-class limb diversity [12]. This refers to the subtle yet vital variations in limb movements that differentiate how individuals perform the same motion, such as walking or running. These limb-specific differences are essential for creating the unique

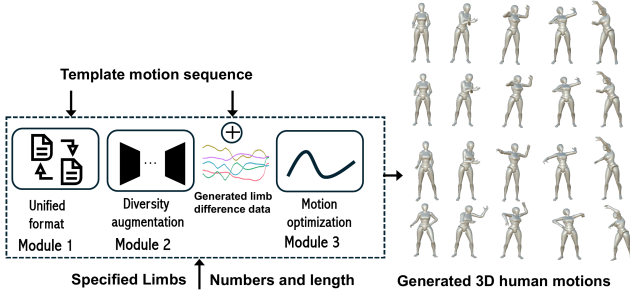


Figure 2: Three modules employed to generate the 3D motion based on template motion sequence.

personalities and customization of characters with unique personalities, but are often inadequately modeled by existing generation or augmentation techniques [9].

To address this specific gap, we present a novel motion generation pipeline that gives more controllable elements during the motion generation process, focusing on enhancing intra-class limb diversity. Rather than generating motions from text or video, it starts with a single motion sequence template, leverages learned patterns to augment limb movements [2], and employs an optimization module to ensure physical plausibility. Based on the conditional Variational Autoencoder (cVAE) structure, it is able to affect the specified limb’s movements and control the number and length of movements generated. This low-cost, controllable solution enables creators to rapidly expand motion libraries with diverse, personalized content, while its main advantages include that users can customize the types of limbs that need to be enhanced, perform limb enhancement for input motions, and generate new motion sequences rapidly and in large numbers.

2 Implementation

Our pipeline can control the changes in motions based on the user-provided labels of the body parts. It could generate multiple variations from a single motion template, and each corresponding body part has internal differences. It consists of three main parts: format unification, diversity augmentation, and motion optimization (Fig.2). When a user uploads a motion sequence, it is first standardized into a unified format compatible with our tool, using the SMPL model [6] as a common skeleton. Then, the core augmentation engine generates new limb movements. Finally, a lightweight optimization pass refines the output to ensure it is visually plausible and natural.

The core of our tool is a generative module designed to enhance intra-class limb diversity. Based on our analysis that most individual motion differences occur in the limbs (especially arms), we trained a Transformer-based cVAE [8] on the HumanAct12 dataset [2]. This model learns the natural variation of limb movements. During generation, the model outputs only the difference data for a user-specified limb, which is then added to the original template motion sequence.

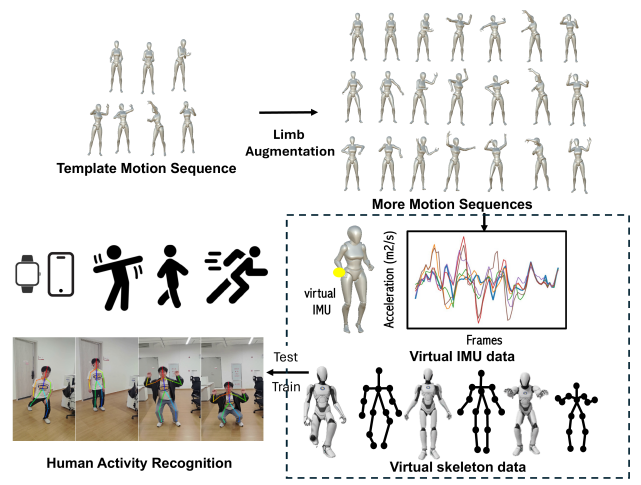


Figure 3: Application of Dataset Synthesis for HAR training including the virtual IMU data and skeleton data.

To ensure the quality of the generated motion, we designed a lightweight optimization module that performs several key functions without the high computational cost of full physical simulation. It uses inverse kinematics to maintain the original motion’s semantic meaning [3], limit joint angles to prevent unnatural poses of motion, applies a smoothing filter to remove jitter, and dynamically balances similarity and variability to preserve the newly generated motion diversity.

3 Application of Dataset Synthesis

Beyond traditional animation content creation, we argue the proposed pipeline is particularly well-suited for synthetic dataset generation—a critical need in fields like human activity recognition (HAR) and embodied intelligence, where access to large-scale, diverse human motion data is often constrained by high collection costs. Below, we demonstrate some scenarios in HAR (Figure 3):

Cross-modal Virtual IMU Dataset Generation For wearable human activity recognition (HAR) relying on inertial measurement units (IMUs), data scarcity due to high collection costs is a critical bottleneck [4]. Our pipeline generates diverse 3D motions from a few templates, which can be converted into virtual IMU signals [13]. These signals retain rich intra-class limb variations, improving HAR model performance—for example, generating varied running or warm-up motions to expand IMU datasets [12].

Video-based HAR System Prototyping 2D skeleton datasets for video-based HAR are often limited by small scales and occlusion issues [4]. This pipeline generated 3D motions can be converted into 2D skeleton sequences via tools like OpenPose [1] or MediaPipe [7], supporting low-cost prototyping of vision-based HAR systems without relying on large real-world video datasets.

4 Conclusion

This poster shows a novel 3D motion generation pipeline, addressing the critical limitation of insufficient limb diversity. Comprising

a unified format module, a cVAE-based limb diversity augmentation module, and a motion optimization module, this lightweight framework generates diverse and natural 3D human motions from a single template sequence while preserving motion semantics. Its core value lies in advancing synthetic dataset generation—a pressing need in fields like human activity recognition (HAR) and embodied intelligence, where large-scale, diverse motion data is scarce due to high collection costs.

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